

Type 2 Diabetes Mellitus Electronic Medical Record Case and Control Selection Algorithms

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1 Overview

This document describes the Northwestern University Type 2 diabetes mellitus (T2DM) algorithms for extracting both T2DM cases and T2DM controls from the electronic medical record (EMR). There are two main parts to this document. The first part (Section 2) provides descriptions of the input data elements to be extracted from the EMR, flowcharts, and pseudo-code descriptions of the algorithms. The second part (Section 3) is an installation guide for executable workflows that implement the T2D case and control selection algorithms. These workflows are based on the Konstanz Information Miner (KNIME) data analysis platform.¹

2 Algorithm Descriptions

The case and control selection algorithms require certain patient-level data elements to be extracted from the EMR. This information includes diagnoses, lab results, medication orders, and physician encounter dates. Lists of codes that satisfy various algorithm requirements (including ICD-9 codes, LOINC codes, and RxNorm codes) are provided in tabular form in Appendix A. Additionally, Section 3.1 contains a translation of these data elements into data dictionaries for input into the KNIME workflow implementations.²

¹Questions about the core algorithms should be sent to japacheco@northwestern.edu, while questions about the executable KNIME workflows should be sent to wkt@northwestern.edu.

²See also the T2D study in the eleMAP online tool. This study contains data elements that were used in a T2D genome-wide association study (GWAS), using a patient cohort derived from the EMR-based algorithm described in this document.

2.1 T2DM Case Selection Algorithm Logic

For the T2D case selection algorithm, the following data elements are required:

1. Counts of T1DM ICD-9 code assignment dates by diagnostic source (Table 3)
2. Counts of T2DM ICD-9 code assignment dates by diagnostic source (Table 4)
3. T1DM medications (i.e., Insulin & Symmlin) order or prescription dates – at least the earliest date of Rx (Table 5)
4. T2DM medications order or prescription dates –at least the earliest date of Rx (Table 6)
5. Fasting blood glucose lab values –at least the maximum value (Table 7)
6. Random blood glucose lab values –at least the maximum value (Table 7)
7. HBA1c lab values –at least the maximum value (Table 7)

For the T2D case selection algorithm, the following definitions apply:

- *Abnormal lab* – An abnormal lab value is defined as one of the following:
 - Random glucose > 200 mg/dl
 - Fasting glucose ≥ 125 mg/dl
 - Hemoglobin A1c $\geq 6.5\%$
- *Physician entered diagnosis* – A physician entered diagnosis code is one that is derived from encounter or problem list sources only (excludes billing codes).

A flowchart expressing the logic of the T2DM case selection algorithm is shown in Figure 1. There are five paths through this flowchart, and these five paths are translated into equivalent pseudo-code in Algorithm 1. This algorithm shows the top-level logic, with additional sub-procedures implementing the lower-level details. These sub-procedures are also expressed below, in terms of an SQL-like syntax that is linked to the ICD-9, LOINC, and RxNorm codes in Appendix A.³

³These are the abbreviations used in the following flowcharts and algorithms:

- DX $\rightarrow$ DIAGNOSIS
- RX $\rightarrow$ PRESCRIPTION
- PT $\rightarrow$ PATIENT
- DT $\rightarrow$ DATE
- CNT $\rightarrow$ COUNT

2.2 T2DM Control Selection Algorithm Logic

For the T2D control selection algorithm, the following data elements are required:

1. Counts of ICD-9 codes related to diabetes (Table 9)
2. Fasting blood glucose lab values –at least the maximum value (Table 7)
3. Random blood glucose lab values –at least the maximum value (Table 7)
4. HBA1c lab values –at least the maximum value (Table 7)
5. Diabetes family history –could be supplemented with self-reported data from a questionnaire
6. T1DM medications (i.e., Insulin & Symlin) order or prescription dates – at least the earliest date of Rx (Table 5)
7. T2DM medications order or prescription dates –at least the earliest date of Rx (Table 6)
8. Diabetes medical supply orders (Table 8)
9. Count of dates the patient had face-to-face outpatient clinic encounters (in-person visits with a clinician)

For the control selection algorithm, the following definition applies:

- *Abnormal lab:* abnormal lab values include:

Random glucose > 110 mg/dl

Fasting glucose $\geq$ 110 mg/dl

Hemoglobin A1c $\geq$ 6.0%

A flowchart expressing the logic of the T2DM control selection algorithm is shown in Figure 2. There is only one path through this flowchart, and this path is translated into equivalent pseudo-code in Algorithm 8. This algorithm shows the top-level logic, with additional sub-procedures implementing the lower-level details. These sub-procedures are also expressed below, in terms of an SQL-like syntax that is linked to the ICD-9, LOINC, and RxNorm codes in Appendix A.

3 KNIME workflow

This section describes installation of executable workflows that implement the case and control algorithms described in Section 2. These workflows are executed inside of the Konstanz Information Miner (KNIME) data analysis platform. The workflows take as input comma-separated value (csv) files, with each row corresponding to a patient (for examples, see the sample input files `dm_potential_cases.csv` and `dm_potential_controls.csv`).

3.1 Data Dictionaries

Each row of input data consists of a set of patient-level variables. We present here the data dictionaries that describe these patient-level input variables. The columns of the dictionaries specify each variable's name, type, and range of possible values. The next column specifies whether or not missing values are permitted, and if so, what the default value of the variable is.⁴ The final column refers (where appropriate) to the table in Appendix A where corresponding code values for the variable can be found.

Table 1: Input variables to the T2DM case selection KNIME workflow

Name	Type	Range	Missing (def.)	Ref.
<i>pat_id</i> (unique)	INTEGER	$n \geq 1$	FALSE	NA
<i>t1dm_dx_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 3
<i>t2dm_dx_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 4
<i>t2dm_physcn_dx_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 4
<i>t1dm_rx_dt</i>	STRING	yyyy-mm-dd	TRUE (NULL)	Table 5
<i>t2dm_rx_dt</i>	STRING	yyyy-mm-dd	TRUE (NULL)	Table 6
<i>max_fast_gluc_lab_val</i>	FLOAT	$n \geq 0 > 0$	TRUE (NULL)	Table 7
<i>max_rndm_gluc_lab_val</i>	FLOAT	$n \geq 0 > 0$	TRUE (NULL)	Table 7
<i>max_hba1c_lab_val</i>	FLOAT (%)	$100.0 \geq n \geq 0.0$	TRUE (NULL)	Table 7

⁴The default value is automatically inserted for a variable when it is missing a specified value.

Table 2: Input variables to the T2DM control selection KNIME workflow

Name	Type	Range	Missing (def.)	Ref.
<i>pat_id</i> (unique)	INTEGER	$n \geq 1$	FALSE	NA
<i>fam_hist_of_dm</i>	INTEGER	$n \in \{0, 1\}$	TRUE (0)	NA
<i>enctrs_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 3
<i>max_fast_gluc_lab_val</i>	FLOAT	$n \geq 0.0$	TRUE (NULL)	Table 7
<i>max_rndm_gluc_lab_val</i>	FLOAT	$n \geq 0.0$	TRUE (NULL)	Table 7
<i>max_hba1c_lab_val</i>	FLOAT (%)	$100.0 \geq n \geq 0.0$	TRUE (NULL)	Table 7
<i>dm_dx_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 9
<i>dm_med_supplies_cnt</i>	INTEGER	$n \geq 0$	TRUE (0)	Table 5, Table 6, Table 8

3.2 Installation and Execution

1. Download and install KNIME (version 2.4 or later). The KNIME website contains installation instructions, as well as tutorials.
2. Download the T2D case and control workflows, which are contained in a single zip file: `T2D-workflows.zip`. Don't unzip the file.
3. Download the two sample input files for the workflows: `dm_potential_cases.csv` and `dm_potential_controls.csv`.
4. Start KNIME. On start-up, you will see an empty workspace similar to the screenshot in Figure 3.
5. Select **File** ⇒ **Import KNIME workflow...** The resulting pop-up window is shown in Figure 4. Click on the **Select archive file:** radio button, and navigate to your local copy of the `T2D-workflows.zip` file. Click on the **Finish** button.
6. Double-click on the **Diabetes-Case-Assignment** workflow to open it. Your workspace will now look similar to the screenshot in Figure 5.⁵
7. Double-click on the **File Reader** node in the workflow graph. You will see the pop-up window shown in Figure 6. Click on the **Browse...** button and navigate to your local copy of the `dm_potential_cases.csv` file. Your pop-up window should look like the one in Figure 7. Make sure that the **read row IDs** box is unchecked, while the **read column headers** box is checked. Click on the **OK** button to close the window.
8. Double-click on the **CSV Writer** node in the workflow graph. Click on the **Browse...** button and navigate to a directory of your choosing where the output file `dm_cases.csv` will be generated. Click on the **OK** button to close the window.
9. The workflow is now ready to execute. Click on the green button with the double arrow at the toolbar at the top, or enter **Shift+F7** on the keyboard. If the nodes of the workflow have already been executed⁶, then first select all nodes (**Control+A**), right click, and select **Reset**.
 - (a) The output file will be located in the directory that you chose in Step 8.
 - (b) Right click on the **Rule Engine** node in the graphical view and select **Classified Data** in order to view the output of the algorithm (Figure 8).
 - (c) Right click on the **Histogram** node in the graphical view and select **View: Histogram View** to get counts of the assignments that were made (Figure 9).

⁵All following steps apply also to the **Diabetes-Control-Assignment** workflow.

⁶An executed node will have a green indicator underneath it.

A DataElements

Table 3: T1DM diagnosis codes. Used in Algorithm 1.

Description	ICD-9 code
Type 1 Diabetes	250.x1, 250.x3

Table 4: T2DM diagnosis codes. Used in Algorithm 1.

Description	ICD-9 code
Type 2 Diabetes	250.x0, 250.x2 (excl. 250.10, 250.12)

Table 5: T1DM medications. Used in Algorithm 1 and Algorithm 8.

Generic Name	Example Brand	RxNorm CUI (ingredient-level)
insulin		139825, 274783, 314684, 352385, 400008, 51428, 5856, 86009
pramlintide	Symlin	139953

Table 6: T2DM medications. Used in Algorithm 1 and Algorithm 8.

Generic Name	Example Brand	RxNorm CUI (ingredient-level)
acetoexamide	Dymelor	173
tolazamide	Tolinase	10633
chlorpropamide	Diabinese	2404
glipizide	Glucotrol	4821
glipizide	Glucotrol XL	217360
glyburide	Micronase, Glynase, Diabeta	4815
glimepiride	Amaryl	25789
repaglinide	Prandin	73044
nateglinide	Starlix	274332
metformin	Glucophage	6809
rosiglitazone	Avandia	84108
pioglitazone	ACTOS	33738
troglitazone	Rezulin	72610
acarbose	Precose	16681
miglitol	Glyset	30009
sitagliptin	Januvia	593411
exenatide	Byetta	60548

Table 7: Diabetes mellitus lab codes. Used in Algorithm 1 and Algorithm 8

Description	LOINC code
Fasting glucose	1558-6
Random glucose	2339-0, 2345-7
Hemoglobin A1C	4548-4, 17856-6, 4549-2, 17855-8

Table 8: Diabetes medical supplies. Used in Algorithm 8.

Description	Source Vocab.	RxNorm CUI (ingredient-level)
Blood-glucose meters & sensors	NDDF	126958, 412956, 412959, 637321, 668291, 668370, 686655, 692383, 748611, 880998, 881056
	VANDF	751128
Insulin syringes	RxNorm	847187, 847191, 847197, 847203, 847207, 847211, 847230, 847239, 847252, 847256, 847259, 847263, 847278, 847416, 847417
	NDDF	806905, 806903, 408119

Table 9: Diabetes mellitus diagnosis codes. Used in Algorithm 8.

Description	ICD-9 code
Diabetes mellitus (T1 & T2)	250.xx
Impaired fasting glucose	790.21
Impaired oral glucose tolerance test	790.22
Abnormal glucose not otherwise spec.	790.2, 790.29
Abnormal glucose during pregnancy	648.8x
Gestational diabetes	648.0x
Glycosuria	791.5
Dysmetabolic syndrome X	277.7
Family history of diabetes mellitus	V18.0
Screening for diabetes mellitus	V77.1